

Essays on the theory of numbers

Richard Dedekind

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Preface

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Transcriber's notes

In this edition, the symbol $\mathfrak{Z}$ is used as a typographical approximation for the author's original *part* of relational symbol, which should not be confused with the numerical digit 3. Correspondingly, the symbol $\mathfrak{E}$ serves as its inverse (*whole* of), representing the converse relationship such that $S \mathfrak{E} A$ is logically equivalent to $A \mathfrak{Z} S$.

Note: Pending the parts about internal page reference and typographical errors.

Part I.

Continuity and irrational numbers

My attention was first directed toward the considerations which form the subject of this pamphlet in the autumn of 1858. As professor in the Polytechnic School in Zürich I found myself for the first time obliged to lecture upon the elements of the differential calculus and felt more keenly than ever before the lack of a really scientific foundation for arithmetic. In discussing the notion of the approach of a variable magnitude to a fixed limiting value, and especially in proving the theorem that every magnitude which grows continually, but not beyond all limits, must certainly approach a limiting value, I had recourse to geometric evidences. Even now such resort to geometric intuition in a first presentation of the differential calculus, I regard as exceedingly useful, from the didactic standpoint, and indeed indispensable, if one does not wish to lose too much time. But that this form of introduction into the differential calculus can make no claim to being scientific, no one will deny. For myself this feeling of dissatisfaction was so overpowering that I made the fixed resolve to keep meditating on the question till I should find a purely arithmetic and perfectly rigorous foundation for the principles of infinitesimal analysis. The statement is so frequently made that the differential calculus deals with continuous magnitude, and yet an explanation of this continuity is nowhere given; even the most rigorous expositions of the differential calculus do not base their proofs upon continuity but, with more or less consciousness of the fact, they either appeal to geometric notions or those suggested by geometry, or depend upon theorems which are never established in a purely arithmetic manner. Among these, for example, belongs the above-mentioned theorem, and a more careful investigation convinced me that this theorem, or any one equivalent to it, can be regarded in some way as a sufficient basis for infinitesimal analysis. It then only remained to discover its true origin in the elements of arithmetic and thus at the same time to secure a real definition of the essence of continuity. I succeeded Nov. 24, 1858, and a few days afterward I communicated the results of my meditations to my dear friend Durège with whom I had a long and lively discussion. Later I explained these views of a scientific basis of arithmetic to a few of my pupils, and here in Braunschweig read a paper upon the subject before the scientific club of professors, but I could not make up my mind to its publication, because, in the first place, the presentation did not seem altogether simple, and further, the theory itself had little promise.

Nevertheless I had already half determined to select this theme as subject for this occasion, when a few days ago, March 14, by the kindness of the author, the paper *Die Elemente der Funktionenlehre* by E. Heine (*Crelle's Journal*, Vol. 74) [1] came into my hands and confirmed me in my decision. In the main I fully agree with the substance of this memoir, and indeed I could hardly do otherwise, but I will frankly acknowledge that my own presentation seems to me to be simpler in form and to bring out the vital point more clearly. While writing this preface (March 20, 1872), I am just in receipt of the interesting paper *Ueber die Ausdehnung eines Satzes aus der Theorie der trigonometrischen Reihen*, by G. Cantor (*Math. Annalen*, Vol. 5) [2], for which I owe the ingenious author my hearty thanks. As I find on a hasty perusal, the axiom given in Section II. of that paper, aside from the form of presentation, agrees with what I designate in Section 3 as the essence of continuity. But what advantage will be gained by even a purely abstract definition of real numbers of a higher type, I am as yet unable to see, conceiving as I do of the domain of real numbers as complete in itself.

1. Properties of rational numbers

The development of the arithmetic of rational numbers is here presupposed, but still I think it worth while to call attention to certain important matters without discussion, so as to show at the outset the standpoint assumed in what follows. I regard the whole of arithmetic as a necessary, or at least natural, consequence of the simplest arithmetic act, that of counting, and counting itself as nothing else than the successive creation of the infinite series of positive integers in which each individual is defined by the one immediately preceding; the simplest act is the passing from an already-formed individual to the consecutive new one to be formed. The chain of these numbers forms in itself an exceedingly useful instrument for the human mind; it presents an inexhaustible wealth of remarkable laws obtained by the introduction of the four fundamental operations of arithmetic. Addition is the combination of any arbitrary repetitions of the above-mentioned simplest act into a single act; from it in a similar way arises multiplication. While the performance of these two operations is always possible, that of the inverse operations, subtraction and division, proves to be limited. Whatever the immediate occasion may have been, whatever comparisons or analogies with experience, or intuition, may have led thereto; it is certainly true that just this limitation in performing the indirect operations has in each case been the real motive for a new creative act; thus negative and fractional numbers have been created by the human mind; and in the system of all rational numbers there has been gained an instrument of infinitely greater perfection. This system, which I shall denote by R , possesses first of all a completeness and self-containedness which I have designated in another place¹ as characteristic of a *body of numbers* [Zahlkörper] and which consists

¹*Vorlesungen über Zahlentheorie*, by P. G. Lejeune Dirichlet. 2d ed. §159 [3].

1. Properties of rational numbers

in this that the four fundamental operations are always performable with any two individuals in R , i.e., the result is always an individual of R , the single case of division by the number zero being excepted.

For our immediate purpose, however, another property of the system R is still more important; it may be expressed by saying that the system R forms a well-arranged domain of one dimension extending to infinity on two opposite sides. What is meant by this is sufficiently indicated by my use of expressions borrowed from geometric ideas; but just for this reason it will be necessary to bring out clearly the corresponding purely arithmetic properties in order to avoid even the appearance as if arithmetic were in need of ideas foreign to it.

To express that the symbols a and b represent one and the same rational number we put $a = b$ as well as $b = a$. The fact that two rational numbers a , b are different appears in this that the difference $a - b$ has either a positive or negative value. In the former case a is said to be *greater* than b , b *less* than a ; this is also indicated by the symbols $a > b$, $b < a$.² As in the latter case $b - a$ has a positive value it follows that $b > a$, $a < b$. In regard to these two ways in which two numbers may differ the following laws will hold:

I. If $a > b$, and $b > c$, then $a > c$. Whenever a , c are two different (or unequal) numbers, and b is greater than the one and less than the other, we shall, without hesitation because of the suggestion of geometric ideas, express this briefly by saying: b lies between the two numbers a , c .

II. If a , c are two different numbers, there are infinitely many different numbers lying between a , c .

III. If a is any definite number, then all numbers of the system R fall into two classes, A_1 and A_2 , each of which contains infinitely many individuals; the first class A_1 comprises all numbers a_1 that are $< a$, the second class A_2 comprises all numbers a_2 that are $> a$; the number a itself may be assigned at pleasure to the first or second class, being respectively the greatest number of the first class or the least of the second. In every case the separation

²Hence in what follows the so-called “algebraic” greater and less are understood unless the word “absolute” is added.

of the system R into the two classes A_1, A_2 is such that every number of the first class A_1 is less than every number of the second class A_2 .

2. Comparison of the rational numbers with the points of a straight line

The above-mentioned properties of rational numbers recall the corresponding relations of position of the points of a straight line L . If the two opposite directions existing upon it are distinguished by “right” and “left,” and p , q are two different points, then either p lies to the right of q , and at the same time q to the left of p , or conversely q lies to the right of p and at the same time p to the left of q . A third case is impossible, if p , q are actually different points. In regard to this difference in position the following laws hold:

I. If p lies to the right of q , and q to the right of r , then p lies to the right of r ; and we say that q lies between the points p and r .

II. If p , r are two different points, then there always exist infinitely many points that lie between p and r .

III. If p is a definite point in L , then all points in L fall into two classes, P_1 , P_2 , each of which contains infinitely many individuals; the first class P_1 contains all the points p_1 , that lie to the left of p , and the second class P_2 contains all the points p_2 that lie to the right of p ; the point p itself may be assigned at pleasure to the first or second class. In every case the separation of the straight line L into the two classes or portions P_1 , P_2 , is of such a character that every point of the first class P_1 lies to the left of every point of the second class P_2 .

2. *Comparison of the rational numbers with the points of a straight line*

This analogy between rational numbers and the points of a straight line, as is well known, becomes a real correspondence when we select upon the straight line a definite origin or zero-point o and a definite unit of length for the measurement of segments. With the aid of the latter to every rational number a a corresponding length can be constructed and if we lay this off upon the straight line to the right or left of o according as a is positive or negative, we obtain a definite end-point p , which may be regarded as the point corresponding to the number a ; to the rational number zero corresponds the point o . In this way to every rational number a , i.e., to every individual in R , corresponds one and only one point p , i.e., an individual in L . To the two numbers a , b respectively correspond the two points, p , q , and if $a > b$, then p lies to the right of q . To the laws I., II., III. of the previous Section correspond completely the laws I., II., III. of the present.

3. Continuity of the straight line

Of the greatest importance, however, is the fact that in the straight line L there are infinitely many points which correspond to no rational number. If the point p corresponds to the rational number a , then, as is well known, the length op is commensurable with the invariable unit of measure used in the construction, i.e., there exists a third length, a so-called common measure, of which these two lengths are integral multiples. But the ancient Greeks already knew and had demonstrated that there are lengths incommensurable with a given unit of length, e.g., the diagonal of the square whose side is the unit of length. If we lay off such a length from the point o upon the line we obtain an end-point which corresponds to no rational number. Since further it can be easily shown that there are infinitely many lengths which are incommensurable with the unit of length, we may affirm: The straight line L is infinitely richer in point-individuals than the domain R of rational numbers in number-individuals.

If now, as is our desire, we try to follow up arithmetically all phenomena in the straight line, the domain of rational numbers is insufficient and it becomes absolutely necessary that the instrument R constructed by the creation of the rational numbers be essentially improved by the creation of new numbers such that the domain of numbers shall gain the same completeness, or as we may say at once, the same *continuity*, as the straight line.

The previous considerations are so familiar and well known to all that many will regard their repetition quite superfluous. Still I regarded this recapitulation as necessary to prepare properly for the main question. For, the way in which the irrational numbers are usually introduced is based directly upon the conception of extensive magnitudes—which itself is nowhere carefully

3. Continuity of the straight line

defined—and explains number as the result of measuring such a magnitude by another of the same kind.¹ Instead of this I demand that arithmetic shall be developed out of itself.

That such comparisons with non-arithmetic notions have furnished the immediate occasion for the extension of the number-concept may, in a general way, be granted (though this was certainly not the case in the introduction of complex numbers); but this surely is no sufficient ground for introducing these foreign notions into arithmetic, the science of numbers. Just as negative and fractional rational numbers are formed by a new creation, and as the laws of operating with these numbers must and can be reduced to the laws of operating with positive integers, so we must endeavor completely to define irrational numbers by means of the rational numbers alone. The question only remains how to do this.

The above comparison of the domain R of rational numbers with a straight line has led to the recognition of the existence of gaps, of a certain incompleteness or discontinuity of the former, while we ascribe to the straight line completeness, absence of gaps, or continuity. In what then does this continuity consist? Everything must depend on the answer to this question, and only through it shall we obtain a scientific basis for the investigation of *all* continuous domains. By vague remarks upon the unbroken connection in the smallest parts obviously nothing is gained; the problem is to indicate a precise characteristic of continuity that can serve as the basis for valid deductions. For a long time I pondered over this in vain, but finally I found what I was seeking. This discovery will, perhaps, be differently estimated by different people; the majority may find its substance very commonplace. It consists of the following. In the preceding section attention was called to the fact that every point p of the straight line produces a separation of the same into two portions such that every point of one portion lies to the left of every point of the other. I find the essence of continuity in the converse, i.e., in the following principle:

¹The apparent advantage of the generality of this definition of number disappears as soon as we consider complex numbers. According to my view, on the other hand, the notion of the ratio between two numbers of the same kind can be clearly developed only after the introduction of irrational numbers.

“If all points of the straight line fall into two classes such that every point of the first class lies to the left of every point of the second class, then there exists one and only one point which produces this division of all points into two classes, this severing of the straight line into two portions.”

As already said I think I shall not err in assuming that every one will at once grant the truth of this statement; the majority of my readers will be very much disappointed in learning that by this commonplace remark the secret of continuity is to be revealed. To this I may say that I am glad if every one finds the above principle so obvious and so in harmony with his own ideas of a line; for I am utterly unable to adduce any proof of its correctness, nor has any one the power. The assumption of this property of the line is nothing else than an axiom by which we attribute to the line its continuity, by which we find continuity in the line. If space has at all a real existence it is *not* necessary for it to be continuous; many of its properties would remain the same even were it discontinuous. And if we knew for certain that space was discontinuous there would be nothing to prevent us, in case we so desired, from filling up its gaps, in thought, and thus making it continuous; this filling up would consist in a creation of new point-individuals and would have to be effected in accordance with the above principle.

4. Creation of irrational numbers

From the last remarks it is sufficiently obvious how the discontinuous domain R of rational numbers may be rendered complete so as to form a continuous domain. In Section 1 it was pointed out that every rational number a effects a separation of the system R into two classes such that every number a_1 of the first class A_1 is less than every number a_2 of the second class A_2 ; the number a is either the greatest number of the class A_1 or the least number of the class A_2 . If now any separation of the system R into two classes A_1, A_2 is given which possesses only *this* characteristic property that every number a_1 in A_1 is less than every number a_2 in A_2 , then for brevity we shall call such a separation a *cut* [Schnitt] and designate it by (A_1, A_2) . We can then say that every rational number a produces one cut or, strictly speaking, two cuts, which, however, we shall not look upon as essentially different; this cut possesses, *besides*, the property that either among the numbers of the first class there exists a greatest or among the numbers of the second class a least number. And conversely, if a cut possesses this property, then it is produced by this greatest or least rational number.

But it is easy to show that there exist infinitely many cuts not produced by rational numbers. The following example suggests itself most readily.

Let D be a positive integer but not the square of an integer, then there exists a positive integer λ such that

$$\lambda^2 < D < (\lambda + 1)^2.$$

4. Creation of irrational numbers

If we assign to the second class A_2 , every positive rational number a_2 whose square is $> D$, to the first class A_1 all other rational numbers a_1 , this separation forms a cut (A_1, A_2) , i.e., every number a_1 is less than every number a_2 . For if $a_1 = 0$, or is negative, then on that ground a_1 is less than any number a_2 , because, by definition, this last is positive; if a_1 is positive, then is its square $\leq D$, and hence a_1 is less than any positive number a_2 whose square is $> D$.

But this cut is produced by no rational number. To demonstrate this it must be shown first of all that there exists no rational number whose square $= D$. Although this is known from the first elements of the theory of numbers, still the following indirect proof may find place here. If there exist a rational number whose square $= D$, then there exist two positive integers t, u , that satisfy the equation

$$t^2 - Du^2 = 0,$$

and we may assume that u is the *least* positive integer possessing the property that its square, by multiplication by D , may be converted into the square of an integer t . Since evidently

$$\lambda u < t < (\lambda + 1)u,$$

the number $u' = t - \lambda u$ is a positive integer certainly *less* than u . If further we put

$$t' = Du - \lambda t,$$

t' is likewise a positive integer, and we have

$$t'^2 - Du'^2 = (\lambda^2 - D)(t^2 - Du^2) = 0,$$

which is contrary to the assumption respecting u .

Hence the square of every rational number x is either $< D$ or $> D$. From this it easily follows that there is neither in the class A_1 a greatest, nor in the class A_2 a least number. For if we put

$$y = \frac{x(x^2 + 3D)}{3x^2 + D},$$

we have

$$y - x = \frac{2x(D - x^2)}{3x^2 + D}$$

and

$$y^2 - D = \frac{(x^2 - D)^3}{(3x^2 + D)^2}.$$

If in this we assume x to be a positive number from the class A_1 , then $x^2 < D$, and hence $y > x$ and $y^2 < D$. Therefore y likewise belongs to the class A_1 . But if we assume x to be a number from the class A_2 , then $x^2 > D$, and hence $y < x$, $y > 0$, and $y^2 > D$. Therefore y likewise belongs to the class A_2 . This cut is therefore produced by no rational number.

In this property that not all cuts are produced by rational numbers consists the incompleteness or discontinuity of the domain R of all rational numbers.

Whenever, then, we have to do with a cut (A_1, A_2) produced by no rational number, we create a new, an *irrational* number α , which we regard as completely defined by this cut (A_1, A_2) ; we shall say that the number α corresponds to this cut, or that it produces this cut. From now on, therefore, to every definite cut there corresponds a definite rational or irrational number, and we regard two numbers as *different* or *unequal* always and only when they correspond to essentially different cuts.

In order to obtain a basis for the orderly arrangement of all *real*, i.e., of all rational and irrational numbers we must investigate the relation between any two cuts (A_1, A_2) and (B_1, B_2) produced by any two numbers α and β . Obviously a cut (A_1, A_2) is given completely when one of the two classes, e.g., the first A_1 is known, because the second A_2 consists of all rational numbers not contained in A_1 , and the characteristic property of such a first class lies in this that if the number a_1 is contained in it, it also contains all

4. Creation of irrational numbers

numbers less than a_1 . If now we compare two such first classes A_1, B_1 with each other, it may happen

1. That they are perfectly identical, i.e., that every number contained in A_1 is also contained in B_1 , and that every number contained in B_1 is also contained in A_1 . In this case A_2 is necessarily identical with B_2 , and the two cuts are perfectly identical, which we denote in symbols by $\alpha = \beta$ or $\beta = \alpha$.

But if the two classes A_1, B_1 are not identical, then there exists in the one, e.g., in A_1 , a number $a'_1 = b'_2$ not contained in the other B_1 and consequently found in B_2 ; hence all numbers b_1 contained in B_1 are certainly less than this number $a'_1 = b'_2$ and therefore all numbers b_1 are contained in A_1 .

2. If now this number a'_1 is the only one in A_1 that is not contained in B_1 , then is every other number a_1 contained in A_1 also contained in B_1 and is consequently $< a'_1$, i.e., a'_1 is the greatest among all the numbers a_1 , hence the cut (A_1, A_2) is produced by the rational number $a = a'_1 = b'_2$. Concerning the other cut (B_1, B_2) we know already that all numbers b_1 in B_1 are also contained in A_1 and are less than the number $a'_1 = b'_2$ which is contained in B_2 ; every other number b_2 contained in B_2 must, however, be greater than b'_2 , for otherwise it would be less than a'_1 , therefore contained in A_1 and hence in B_1 ; hence b'_2 is the least among all numbers contained in B_2 , and consequently the cut (B_1, B_2) is produced by the same rational number $\beta = b'_2 = a'_1 = \alpha$. The two cuts are then only unessentially different.

3. If, however, there exist in A_1 at least two different numbers $a'_1 = b'_2$ and $a''_1 = b''_2$, which are not contained in B_1 , then there exist infinitely many of them, because all the infinitely many numbers lying between a'_1 and a''_1 are obviously contained in A_1 [Section 1, II.](#) but not in B_1 . In this case we say that the numbers α and β corresponding to these two essentially different cuts (A_1, A_2) and (B_1, B_2) are *different*, and further that α is *greater* than β , that β is *less* than α , which we express in symbols by $\alpha > \beta$ as well as $\beta < \alpha$. It is to be noticed that this definition coincides completely with the one given earlier, when α, β are rational.

The remaining possible cases are these:

4. If there exists in B_1 one and only one number $b'_1 = a'_2$, that is not contained in A_1 then the two cuts (A_1, A_2) and (B_1, B_2) are only unessentially different and they are produced by one and the same rational number $\alpha = a'_2 = b'_1 = \beta$.

5. But if there are in B_1 at least two numbers which are not contained in A_1 , then $\beta > \alpha$, $\alpha < \beta$.

As this exhausts the possible cases, it follows that of two different numbers one is necessarily the greater, the other the less, which gives two possibilities. A third case is impossible. This was indeed involved in the use of the *comparative* (greater, less) to designate the relation between α , β ; but this use has only now been justified. In just such investigations one needs to exercise the greatest care so that even with the best intention to be honest he shall not, through a hasty choice of expressions borrowed from other notions already developed, allow himself to be led into the use of inadmissible transfers from one domain to the other.

If now we consider again somewhat carefully the case $\alpha > \beta$ it is obvious that the less number β , if rational, certainly belongs to the class A_1 ; for since there is in A_1 a number $a'_1 = b'_2$ which belongs to the class B_2 , it follows that the number β , whether the greatest number in B_1 or the least in B_2 is certainly $\leq a'_1$ and hence contained in A_1 . Likewise it is obvious from $\alpha > \beta$ that the greater number α , if rational, certainly belongs to the class B_2 , because $\alpha \geq a'_1$. Combining these two considerations we get the following result: If a cut is produced by the number α then any rational number belongs to the class A_1 or to the class A_2 according as it is less or greater than α ; if the number α is itself rational it may belong to either class.

From this we obtain finally the following: If $\alpha > \beta$, i.e., if there are infinitely many numbers in A_1 not contained in B_1 then there are infinitely many such numbers that at the same time are different from α and from β ; every such rational number c is $< \alpha$, because it is contained in A_1 and at the same time it is $> \beta$ because contained in B_2 .

5. Continuity of the domain of real numbers

In consequence of the distinctions just established the system $\mathfrak{R}$ of all real numbers forms a well-arranged domain of one dimension; this is to mean merely that the following laws prevail:

I. If $\alpha > \beta$, and $\beta > \gamma$, then is also $\alpha > \gamma$. We shall say that the number β lies between α and γ .

II. If α, γ are any two different numbers, then there exist infinitely many different numbers β lying between α, γ .

III. If α is any definite number then all numbers of the system $\mathfrak{R}$ fall into two classes $\mathfrak{A}_1$ and $\mathfrak{A}_2$ each of which contains infinitely many individuals; the first class $\mathfrak{A}_1$ comprises all the numbers α_1 that are less than α , the second $\mathfrak{A}_2$ comprises all the numbers α_2 that are greater than α ; the number α itself may be assigned at pleasure to the first class or to the second, and it is respectively the greatest of the first or the least of the second class. In each case the separation of the system $\mathfrak{R}$ into the two classes $\mathfrak{A}_1, \mathfrak{A}_2$ is such that every number of first class $\mathfrak{A}_1$ is smaller than every number of the second class $\mathfrak{A}_2$ and we say that this separation is produced by the number α .

For brevity and in order not to weary the reader I suppress the proofs of these theorems which follow immediately from the definitions of the previous section.

Beside these properties, however, the domain $\mathfrak{R}$ possesses also *continuity*; i.e., the following theorem is true:

5. Continuity of the domain of real numbers

IV. If the system $\mathfrak{R}$ of all real numbers breaks up into two classes $\mathfrak{A}_1, \mathfrak{A}_2$ such that every number α_1 of the class $\mathfrak{A}_1$ is less than every number α_2 of the class $\mathfrak{A}_2$ then there exists one and only one number α by which this separation is produced.

Proof. By the separation or the cut of $\mathfrak{R}$ into $\mathfrak{A}_1$ and $\mathfrak{A}_2$ we obtain at the same time a cut (A_1, A_2) of the system R of all rational numbers which is defined by this that A_1 contains all rational numbers of the class $\mathfrak{A}_1$ and A_2 all other rational numbers, i.e., all rational numbers of the class $\mathfrak{A}_2$. Let α be the perfectly definite number which produces this cut (A_1, A_2) . If β is any number different from α , there are always infinitely many rational numbers c lying between α and β . If $\beta < \alpha$, then $c < \alpha$; hence c belongs to the class A_1 and consequently also to the class $\mathfrak{A}_1$, and since at the same time $\beta < c$ then β also belongs to the same class $\mathfrak{A}_1$, because every number in $\mathfrak{A}_2$ is greater than every number c in $\mathfrak{A}_1$. But if $\beta > \alpha$, then is $c > \alpha$; hence c belongs to the class A_2 and consequently also to the class $\mathfrak{A}_2$, and since at the same time $\beta > c$, then β also belongs to the same class $\mathfrak{A}_2$, because every number in $\mathfrak{A}_1$ is less than every number c in $\mathfrak{A}_2$. Hence every number β different from α belongs to the class $\mathfrak{A}_1$ or to the class $\mathfrak{A}_2$ according as $\beta < \alpha$ or $\beta > \alpha$; consequently α itself is either the greatest number in $\mathfrak{A}_1$ or the least number in $\mathfrak{A}_2$, i.e., α is one and obviously the only number by which the separation of R into the classes $\mathfrak{A}_1, \mathfrak{A}_2$ is produced. Which was to be proved. $\square$

6. Operations with real numbers

To reduce any operation with two real numbers α, β to operations with rational numbers, it is only necessary from the cuts $(A_1, A_2), (B_1, B_2)$ produced by the numbers α and β in the system R to define the cut (C_1, C_2) which is to correspond to the result of the operation, γ . I confine myself here to the discussion of the simplest case, that of addition.

If c is any rational number, we put it into the class C_1 , provided there are two numbers one a_1 in A_1 and one b_1 in B_1 such that their sum $a_1 + b_1 \geq c$; all other rational numbers shall be put into the class C_2 . This separation of all rational numbers into the two classes C_1, C_2 evidently forms a cut, since every number c_1 in C_1 is less than every number c_2 in C_2 . If both α and β are rational, then every number c_1 contained in C_1 is $\leq \alpha + \beta$, because $a_1 \leq \alpha, b_1 \leq \beta$, and therefore $a_1 + b_1 \leq \alpha + \beta$; further, if there were contained in C_2 a number $c_2 < \alpha + \beta$, hence $\alpha + \beta = c_2 + p$, where p is a positive rational number, then we should have

$$c_2 = (\alpha - \frac{1}{2}p) + (\beta - \frac{1}{2}p),$$

which contradicts the definition of the number c_2 , because $\alpha - \frac{1}{2}p$ is a number in A_1 , and $\beta - \frac{1}{2}p$ a number in B_1 ; consequently every number c_2 contained in C_2 is $\geq \alpha + \beta$. Therefore in this case the cut (C_1, C_2) is produced by the sum $\alpha + \beta$. Thus we shall not violate the definition which holds in the arithmetic of rational numbers if in all cases we understand by the sum $\alpha + \beta$ of any two real numbers α, β that number γ by which the cut (C_1, C_2) is produced. Further, if only one of the two numbers α, β is rational, e.g., α , it is easy to see that it makes no difference with the sum

6. Operations with real numbers

$\gamma = \alpha + \beta$ whether the number α is put into the class A_1 or into the class A_2 .

Just as addition is defined, so can the other operations of the so-called elementary arithmetic be defined, viz., the formation of differences, products, quotients, powers, roots, logarithms, and in this way we arrive at real proofs of theorems (as, e.g., $\sqrt{2} \cdot \sqrt{3} = \sqrt{6}$), which to the best of my knowledge have never been established before. The excessive length that is to be feared in the definitions of the more complicated operations is partly inherent in the nature of the subject but can for the most part be avoided. Very useful in this connection is the notion of an *interval*, i.e., a system A of rational numbers possessing the following characteristic property: if a and a' are numbers of the system A , then are all rational numbers lying between a and a' contained in A . The system R of all rational numbers, and also the two classes of any cut are intervals. If there exist a rational number a_1 which is less and a rational number a_2 which is greater than every number of the interval A , then A is called a finite interval; there then exist infinitely many numbers in the same condition as a_1 and infinitely many in the same condition as a_2 ; the whole domain R breaks up into three parts A_1 , A , A_2 and there enter two perfectly definite rational or irrational numbers α_1 , α_2 which may be called respectively the lower and upper (or the less and greater) *limits* of the interval; the lower limit α_1 is determined by the cut for which the system A_1 forms the first class and the upper α_2 by the cut for which the system A_2 forms the second class. Of every rational or irrational number α lying between α_1 and α_2 it may be said that it lies *within* the interval A . If all numbers of an interval A are also numbers of an interval B , then A is called a portion of B .

Still lengthier considerations seem to loom up when we attempt to adapt the numerous theorems of the arithmetic of rational numbers (as, e.g., the theorem $(a + b)c = ac + bc$) to any real numbers. This, however, is not the case. It is easy to see that it all reduces to showing that the arithmetic operations possess a certain continuity. What I mean by this statement may be expressed in the form of a general theorem:

“If the number λ is the result of an operation performed on

the numbers $\alpha, \beta, \gamma, \dots$ and λ lies within the interval L , then intervals $A, B, C, \dots$ can be taken within which lie the numbers $\alpha, \beta, \gamma, \dots$ such that the result of the same operation in which the numbers $\alpha, \beta, \gamma, \dots$ are replaced by arbitrary numbers of the intervals $A, B, C, \dots$ is always a number lying within the interval L ."

The forbidding clumsiness, however, which marks the statement of such a theorem convinces us that something must be brought in as an aid to expression; this is, in fact, attained in the most satisfactory way by introducing the ideas of *variable magnitudes*, *functions*, *limiting values*, and it would be best to base the definitions of even the simplest arithmetic operations upon these ideas, a matter which, however, cannot be carried further here.

7. Infinitesimal analysis

Here at the close we ought to explain the connection between the preceding investigations and certain fundamental theorems of infinitesimal analysis.

We say that a variable magnitude x which passes through successive definite numerical values approaches a fixed limiting value α when in the course of the process x lies finally between two numbers between which α itself lies, or, what amounts to the same, when the difference $x - \alpha$ taken absolutely becomes finally less than any given value different from zero.

One of the most important theorems may be stated in the following manner:

“If a magnitude x grows continually but not beyond all limits it approaches a limiting value.”

I prove it in the following way. By hypothesis there exists one and hence there exist infinitely many numbers α_2 such that x remains continually $< \alpha_2$; I designate by $\mathfrak{A}_2$ the system of all these numbers α_2 , by $\mathfrak{A}_1$ the system of all other numbers α_1 ; each of the latter possesses the property that in the course of the process x becomes finally $\geq \alpha_1$, hence every number α_1 is less than every number α_2 and consequently there exists a number α which is either the greatest in $\mathfrak{A}_1$ or the least in $\mathfrak{A}_2$ ([Section 5, IV.](#)). The former cannot be the case since x never ceases to grow, hence α is the least number in $\mathfrak{A}_2$. Whatever number α_1 be taken we shall have finally $\alpha_1 < x < \alpha$, i.e., x approaches the limiting value α .

This theorem is equivalent to the principle of continuity, i.e., it loses its validity as soon as we assume a single real number not to be contained in

7. Infinitesimal analysis

the domain $\mathfrak{R}$; or otherwise expressed: if this theorem is correct, then is also theorem [Section 5, IV.](#) correct.

Another theorem of infinitesimal analysis, likewise equivalent to this, which is still oftener employed, may be stated as follows:

“If in the variation of a magnitude x we can for every given positive magnitude δ assign a corresponding position from and after which x changes by less than δ then x approaches a limiting value.”

This converse of the easily demonstrated theorem that every variable magnitude which approaches a limiting value finally changes by less than any given positive magnitude can be derived as well from the preceding theorem as directly from the principle of continuity. I take the latter course. Let δ be any positive magnitude (i.e., $\delta > 0$), then by hypothesis a time will come after which x will change by less than δ , i.e., if at this time x has the value a , then afterwards we shall continually have $x > a - \delta$ and $x < a + \delta$. I now for a moment lay aside the original hypothesis and make use only of the theorem just demonstrated that all later values of the variable x lie between two assignable finite values. Upon this I base a double separation of all real numbers. To the system $\mathfrak{A}_2$ I assign a number α_2 (e.g., $a + \delta$) when in the course of the process x becomes finally $\leq \alpha_2$; to the system $\mathfrak{A}_1$ I assign every number not contained in $\mathfrak{A}_2$; if α_1 is such a number, then, however far the process may have advanced, it will still happen infinitely many times that $x > \alpha_2$. Since every number α_1 is less than every number α_2 there exists a perfectly definite number α which produces this cut $(\mathfrak{A}_1, \mathfrak{A}_2)$ of the system $\mathfrak{R}$ and which I will call the upper limit of the variable x which always remains finite. Likewise as a result of the behavior of the variable x a second cut $(\mathfrak{B}_1, \mathfrak{B}_2)$ of the system $\mathfrak{R}$ is produced; a number β_2 (e.g., $a - \delta$) is assigned to $\mathfrak{B}_2$ when in the course of the process x becomes finally $\geq \beta_2$; every other number β_1 , to be assigned to $\mathfrak{B}_1$, has the property that x is never finally $\geq \beta_1$; therefore infinitely many times x becomes $< \beta_1$; the number β by which this cut is produced I call the lower limiting value of the variable x . The two numbers α, β are obviously characterised by the following property: if ϵ is an arbitrarily small positive magnitude

then we have always finally $x < \alpha + \epsilon$ and $x > \beta - \epsilon$, but never finally $x < \alpha - \epsilon$ and never finally $x > \beta + \epsilon$. Now two cases are possible. If α and β are different from each other, then necessarily $\alpha > \beta$, since continually $\alpha_2 \geq \beta_2$; the variable x oscillates, and, however far the process advances, always undergoes changes whose amount surpasses the value $(\alpha - \beta) - 2\epsilon$ where ϵ is an arbitrarily small positive magnitude. The original hypothesis to which I now return contradicts this consequence; there remains only the second case $\alpha = \beta$ since it has already been shown that, however small be the positive magnitude ϵ , we always have finally $x < \alpha + \epsilon$ and $x > \beta - \epsilon$, x approaches the limiting value α , which was to be proved.

These examples may suffice to bring out the connection between the principle of continuity and infinitesimal analysis.

Part II.

The nature and meaning of numbers

8. Systems of elements

9. Transformation of a system

10. Similarity of a transformation

11. Transformation of a system in itself

12. The finite and infinite

13. Simply infinite systems

14. Greater and less numbers

15. Finite and infinite parts

16. Definition by induction

17. Class of simply infinite systems

18. Addition of numbers

19. Multiplication of numbers

20. Involution of numbers

21. Number of elements of a finite system

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